

The Farmer's Problem

An Introduction to Stochastic Linear Programming

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1 Introduction

Every sowing time, a farmer needs to allocate his land for different crops, to find an optimal solution which could both meet his feeding requirement and bring a lot of profit.

The decision procedure contains two stages, the first is to allocate the land for various crops given that the total area is fixed, and the second stage mainly involves exchanging crops with money. After harvest, any production in excess of the feeding requirement would be sold and the farmer will buy crops whose amount of yields are below his requirement amount. Some random variables — yields, weather, prices, which used to be uncertain at first, become known in the second stage. These variables' actual values have an influence on the decisions made in the second stage.

In the first stage, the final aim is to optimize his land allocation. He has n types of crops to grow, the total area is defined as b , investment for unit area devoted for each crop to grow will be denoted by a vector $c = (c_1, c_2, \dots, c_n)^T \in \mathbb{R}^n$, c_1 is the first type of crop's cost per unit area, etc. Basic feeding requirement is named $h \in \mathbb{R}^n$. The farmer decides the amount of area devoted for each crop, the decision is represented by a vector $x \in \mathbb{R}^n$.

Besides, in the second stage, the final aim is to decide the amount of selling and purchasing for each crop, there are two variables used to be uncertain in the first stage and become known now, of which the decision in this stage is dependent. Yields of them is represented by a random vector $T \in \mathbb{R}^n$, selling and buying prices are represented by a random vector $q \in \mathbb{R}^{2n}$. The farmer decide the amount of purchase and selling, the decision is represented by a vector $y \in \mathbb{R}^{2n}$.

He will need to decide twice, the first decision is x , and the second is y .

At first the first stage decision x is determined. After harvest, all the random variables related to the second stage become certain. Then y could be

determined as the optimal solution of the linear programming problem

$$\begin{aligned} & \max q^T y \\ \text{s.t. } & Wy = h - T^T x, \\ & y \geq 0. \end{aligned} \tag{1}$$

where the entries of the vector $W \in \mathbb{R}^{1 \times 2n}$ are the exchange directions, which could only be -1 if products were bought or 1 if products were sold. We need to find the optimal solution y , so that $q^T y$ will be maximized as $Q = \max q^T y$. Thus, profit could be calculated since all variables become certain after x is determined. Since q, T are random variables, then Q and the highest profit is also a random variable. The farmer's aim is to maximize his profit $P = Q - c^T x$. The distribution of Q is determined by x .

The aim is to maximize $\mathbb{E}(P)$.

$$\begin{aligned} & \max \mathbb{E}(Q) - c^T x \\ \text{s.t. } & Ax = b, \\ & x \geq 0. \end{aligned} \tag{2}$$

where $A = (1, 1, \dots, 1)$. And it is not a linear programming problem because $\mathbb{E}(Q)$ may not be a linear function of x .

Since the value of x directly determine the expectation of the profit, our aim is to determine an optimal decision x to attain the highest profit.

2 Theory

Based on the above, a general problem is formed as follows, with certain solution methods related to it.

A two-stage stochastic linear program is formulated as follows

$$\begin{aligned} & \min \mathbb{E}(q(\omega)^T y(\omega)) + c^T x \\ \text{s.t. } & Ax = b, \\ & Wy(\omega) = h(\omega) - T(\omega)x, \\ & y(\omega) \geq 0, x \geq 0. \end{aligned} \tag{3}$$

where ω is a random variable deciding all the other random variables with respect to this problem, c is a known vector in $\mathbb{R}^{n_1}$, b is a known vector in $\mathbb{R}^{m_1}$. Matrices A and W are also known. They are of size $m_1 \times n_1$ and $m_2 \times n_2$, respectively. The stochastic components $q(\omega) \in \mathbb{R}^{n_2}$, $h(\omega) \in \mathbb{R}^{m_2}$, $T(\omega)$ is $m_2 \times n_1$.

For simplicity, we assume that ω is a discrete random variable. For each possible value of ω , we define

$$K(\omega) = \{x \mid \exists y > 0 \text{ s.t. } Wy = h(\omega) - T(\omega)x\},$$

and for every given ω , we define

$$K = \bigcap_{\omega \in \Omega} K(\omega)$$

where $\Omega \subset R^{n_1 \times m_2 + n_2 + m_2}$ is the set of all the possible values of ω . K is said to be elementary feasible set.

Theorem 2.1.

- a. $K(\omega)$ is a convex polyhedron for any given ω .
- b. K is a convex polyhedron if ω is a finite discrete random variable.

Proof. a. As the elementary feasible set of a given linear program problem is always a convex polyhedron. When ω is given, the general problem we raised above is a linear program problem, so $K(\omega)$ is a convex polyhedron.

b. This is because the intersection of finitely many convex polyhedra is also a convex polyhedron. $\square$

Theorem 2.2. For any given ω , $Q(\omega)$ would be finite if we changed the definition of $Q(\omega)$ by adding a constraint so that y has upper bounds, and if $x \in K$.

Proof. Given that ω is certain, when y has upper bounds, the unboundedness of $Q(\omega)$ could be avoided. And when $x \in K$, the given linear program in (3) must be feasible. $\square$

Remark 2.3. In reality, y represents the amount of purchasing and selling, which could not exceed a given number, due to physical constraints. Hence, we could always impose an upper bound on y .

Theorem 2.4. For any given ω , $Q(\omega)$ is a piecewise linear convex function in (h, T) and in $x \in K$, and is a piecewise linear concave function in q .

Proof. Let $f(b) = \min\{q(\omega)^T y \mid Wy = b, y \geq 0\}$. To prove $f(b)$ is convex in b , for any optimal solution y_1, y_2 , which are the optimal solutions for $f(b_1), f(b_2)$, respectively, and for any $\lambda \in (0, 1)$, we have $\lambda f(b_1) + (1 - \lambda)f(b_2) = q(\omega)\lambda y_1 + q(\omega)(1 - \lambda)y_2 = q(\omega)(\lambda y_1 + (1 - \lambda)y_2)$. Let y_3 be an optimal solution for $f(\lambda b_1 + (1 - \lambda)b_2)$. Then $f(\lambda b_1 + (1 - \lambda)b_2) = q(\omega)y_3 \leq q(\omega)(\lambda y_1 + (1 - \lambda)y_2) = \lambda f(b_1) + (1 - \lambda)f(b_2)$. Thus, convexity is proved. Now let $b = h(\omega) - T(\omega)x$. Clearly, b is linear in (h, T) and x . Therefore, $Q(\omega)$ is convex in (h, T) and x . Similarly, $Q(\omega)$ is concave in q .

To prove $f(b)$ is piecewise linear in b , we only need to prove every vertex of the polyhedron $K = \{y \mid y \geq 0, Wy = b\}$ is linear in b . That's because the optimal solution of a linear program are all the vertex of the feasible set, which could be found by the following method. And if all the vertexes are linear in b , then $f(b)$ is piecewise linear.

The coordinates of vertex could be determined as follows. We add $n_2 - n_1$ rows to W , each of which only contains one non-zero entry which is 1, and denote the new square matrix as W_1 . In the meanwhile, we add the same amount of 0's (as new rows) to b , the new vector is denoted as b_1 . Then the

coordinates of a vertex could be determined by the equation $W_1 y = b_1$, because it contains two parts: firstly, basic constraint $W y = b$. Secondly, the equations $(0, \dots, 1, \dots, 0) y = 0$ represent all the coordinates of the vertex which are supposed to be 0. W_1 is invertible, and vertex could be represented as $y = W_1^{-1} b_1$. Every vertex y we found is linear in b . Thus, $f(b)$ is piecewise linear in b . Therefore, $Q(\omega)$ is piecewise linear in (h, T) and x . Similarly, $Q(\omega)$ is piecewise linear in q . $\square$

Corollary 2.5. *If ω is a finite discrete random variable, then $E(Q)$ is piecewise linear and convex in x .* $\square$

References

- [1] John R. Birge, François Louveaux. *Introduction to Stochastic Programming*. Springer 2011.